EXPLORING QUANTUM-RESISTANT BLOCKCHAIN SYSTEM

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ABSTRACT

Quantum computers will ultimately compromise the cryptographic foundations of today's blockchains. While post-quantum signature schemes have emerged as potential replacements, their practicality in high-throughput production systems remains uncertain. In this work, we modify the MEMO blockchain to evaluate two quantum-resistant cryptographic approaches, ML-DSA-44 and Falcon-512, compare them against the classical Ed25519 scheme, and conduct our evaluation on a 48-core (96-thread) server using ZeroMQ-based messaging on 1 M-transaction workloads. At sustained load all four schemes converge to **18–20K TPS** within 10%: Ed25519 **20,437**, Hybrid **20,007**, Falcon-512 **19,482**, ML-DSA-44 **18,353**. The dominant bottleneck is message queuing rather than cryptographic computation. These findings demonstrate that post-quantum blockchains are not only feasible, but match the performance of current classical designs.

MOTIVATION

- **The quantum threat is unavoidable.** Shor's algorithm will likely be able to break widely deployed elliptic-curve schemes such as Ed25519 and ECDSA with a sufficiently large quantum computer.
- **Post-quantum cryptography suffers from a perception gap.** Larger keys and signatures have led to the assumption that PQC schemes are inherently impractical for blockchains, despite limited evaluation under realistic concurrent workloads.
- **Microbenchmarks fail to capture system behavior.** Traditional single-threaded cryptographic benchmarks ignore the effects of concurrency, queue contention, synchronization overheads, and end-to-end transaction processing pipelines.
- **We evaluate the entire blockchain stack.** We integrate post-quantum cryptography directly into a concurrent blockchain execution pipeline, producing throughput measurements that reflect realistic system-level performance rather than isolated cryptographic operations.

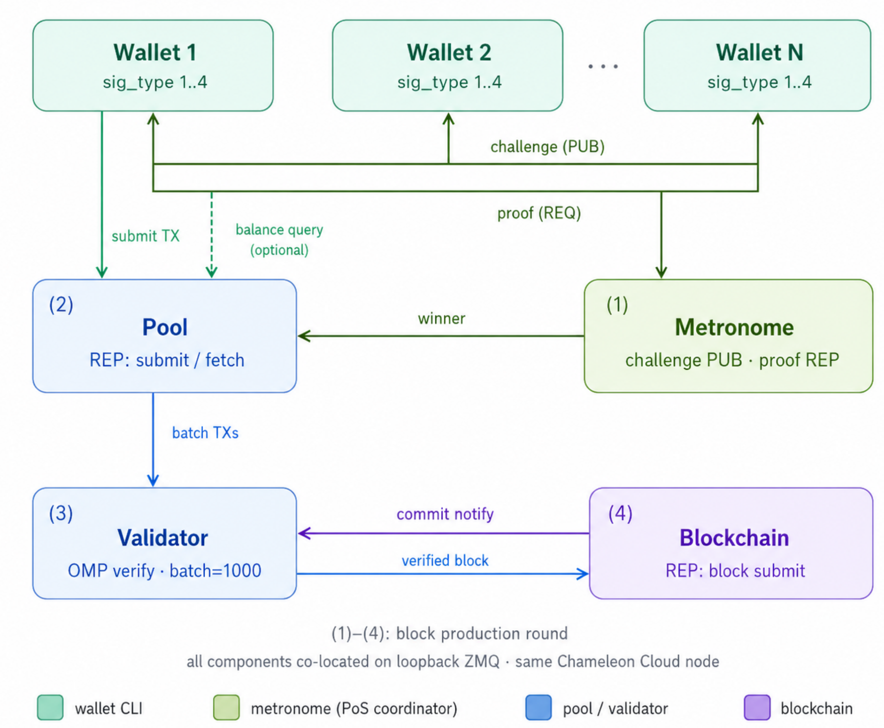
TESTBED

Field	Value
CPU	Intel Xeon Gold 6240R
Topology	48 C / 96 T (2 sockets)
Memory	192 GB DDR4
Storage	NVMe SSD
Compiler	GCC 11 `-O2 -fopenmp`
PQC lib	liboqs 0.11
Classical	OpenSSL 3 EVP

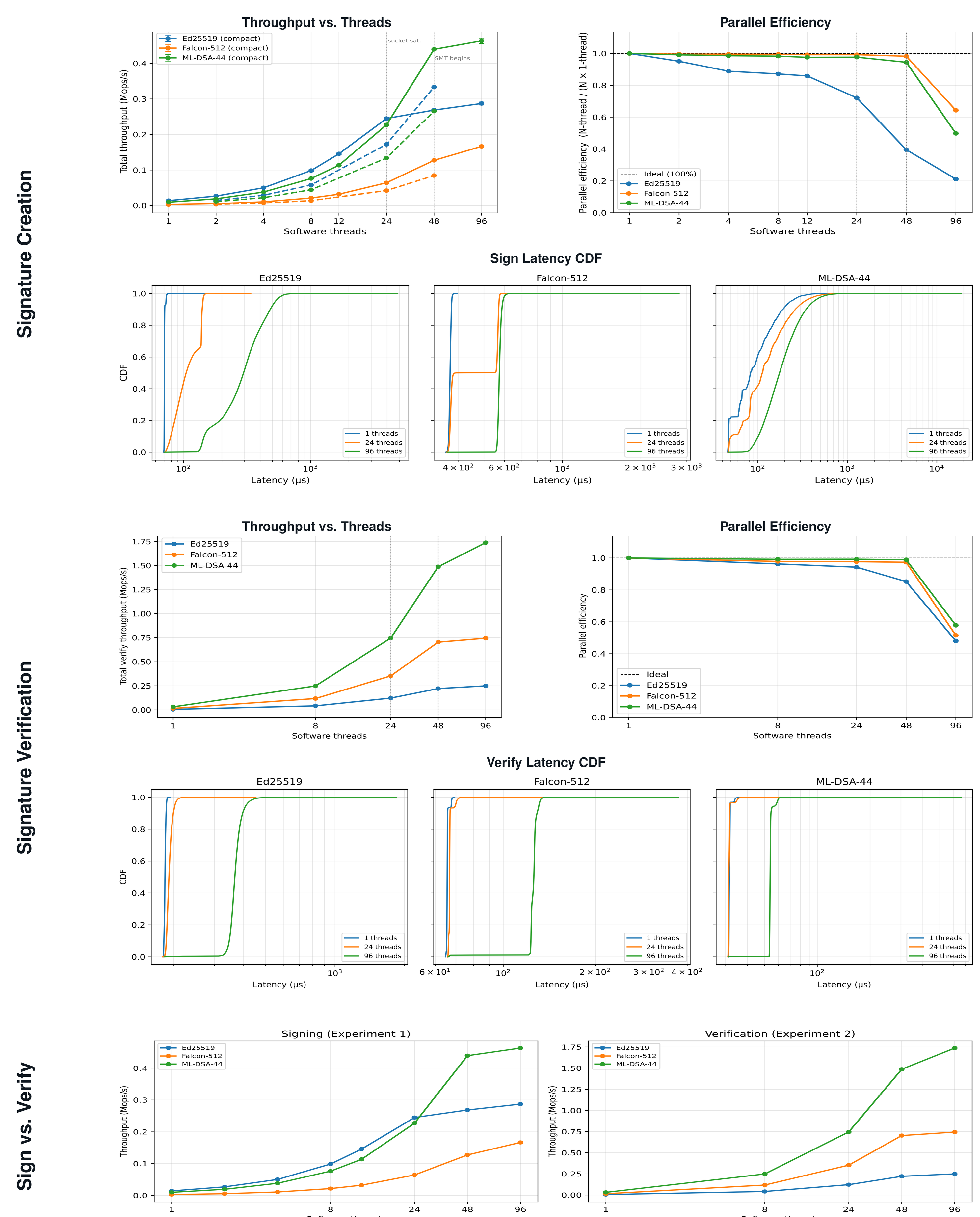
SIGNATURE SCHEMES

Scheme	Pubkey	Sig	Verify	PQC
Ed25519	32 B	64 B	≈60 µs	×
ML-DSA-44	1312 B	2420 B	≈40 µs	✓
Falcon-512	897 B	666 B	≈50 µs	✓
Hybrid	varies	varies	combined	✓

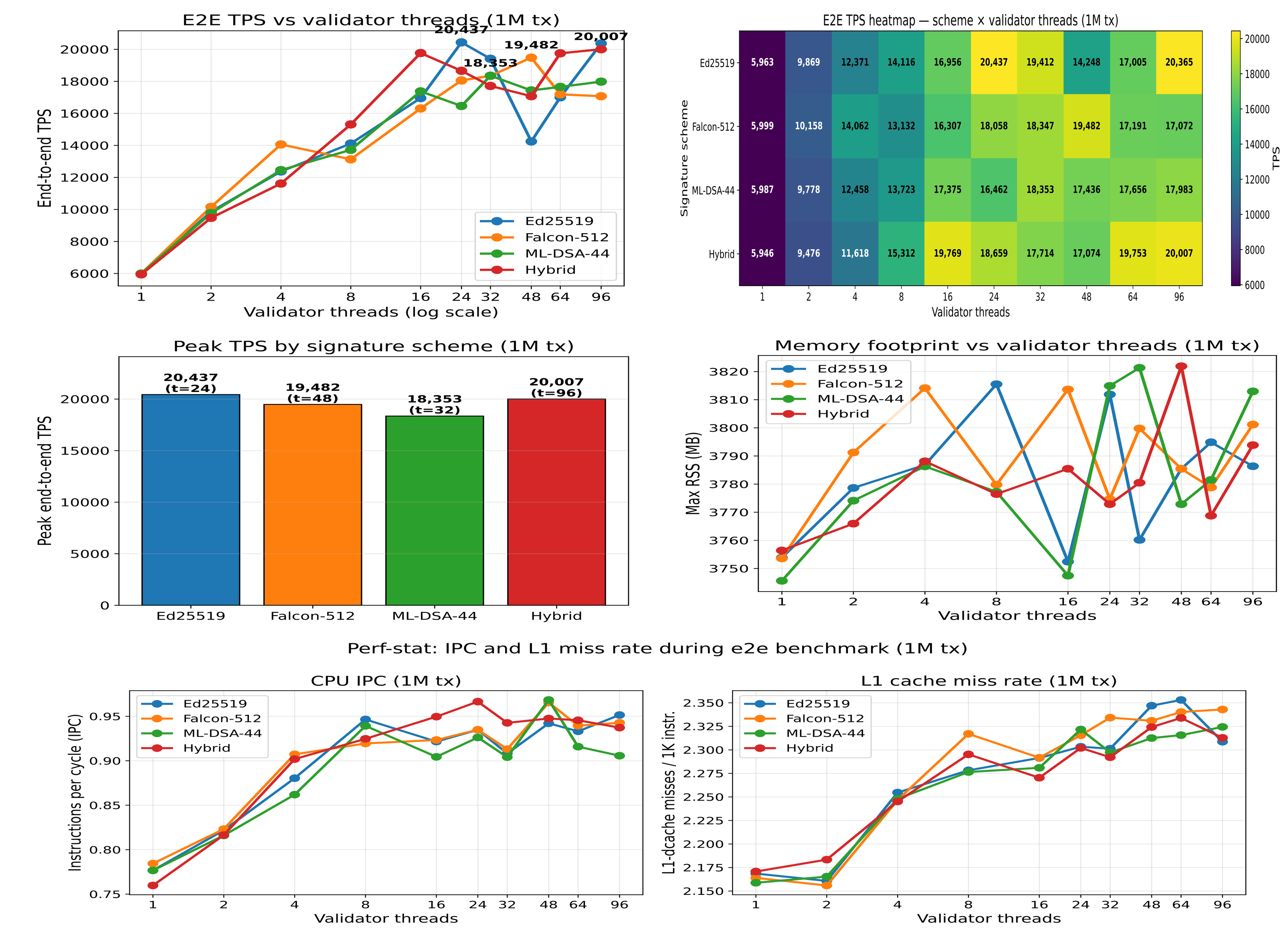
ARCHITECTURE



MICROBENCHMARKING



END-TO-END BLOCKCHAIN TRANSACTION



LIMITATIONS OF CURRENT EXPERIMENTS

- Loopback network only. Real WAN latency will shift absolute TPS numbers.
- Single-node deployment. No distributed consensus tested.
- ZMQ queue is a hard ceiling at ≈18K TPS. Further scaling requires architectural redesign.

CONCLUSION

- **Post-quantum blockchains match classical at scale.** At 1 M sustained load all four schemes converge to 18–20K TPS: Ed25519 20,437, Hybrid 20,007, Falcon-512 19,482, ML-DSA-44 18,353. The ZMQ message bus saturates first, regardless of scheme.
- **Signature size is the real cost.** Sig grows from 64 B (Ed25519) to 666 B (Falcon-512) and 2420 B (ML-DSA-44) – a 10×–37× increase, requiring more network bandwidth per transaction and more permanent storage per block.

Future Work

- Shard the TX pool to break the ZMQ bottleneck and target 50K+ TPS.
- Add SLH-DSA (hash-based, NIST standard) as a fourth standalone PQC scheme.
- Extend to multi-node deployment with real network topology and distributed validator consensus.
- Implement cross-shard atomic commits for partition-safe transaction finality.